

## Quiz

Thursday, February 19, 2026 8:38 AM

- Convert one of the following 16-bit floating point values to decimal:

**Hard Mode (10 pts):**      **C0   CA**

- The register uses IEEE 754 half precision (binary16) with the following number of bits assigned to each component:
  - 1 sign
  - 10 mantissa
  - 5 exponent
- This format uses a bias of 15

$$C0_{16} \Rightarrow 1100\ 0000_2$$

$$CA_{16} \Rightarrow 1100\ 1010_2$$

$$11000000|1001010$$

$$s = 1$$

$$e = 10000 \Rightarrow 16_{10}$$

$$m = 0.0011001010_2 \Rightarrow 1 \times 2^7 + 1 \times 2^6 + 1 \times 2^3 + 1 \times 2^1 \Rightarrow 202$$

$$1 + \frac{202}{1024} \Rightarrow 1 + 0.197265625 = 1.197265625$$

$$(-1)^1 \times 1.197265625 \times 2 = -2.39453125$$